GCSE CHEMISTRY

Higher Tier – Paper 2 style

Topic 4.8: Chemical analysis

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 120 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **112**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 112

0 1 This question is about pure substances and mixtures.

0 1 . 1 What does a chemist mean by a **pure** substance? **[1 mark]**

0 1 . 2 A carton of milk is labelled 'pure milk'. Explain why a chemist would not call milk a pure substance. **[2 marks]**

0 1 . 3 What is a **formulation**? **[2 marks]**

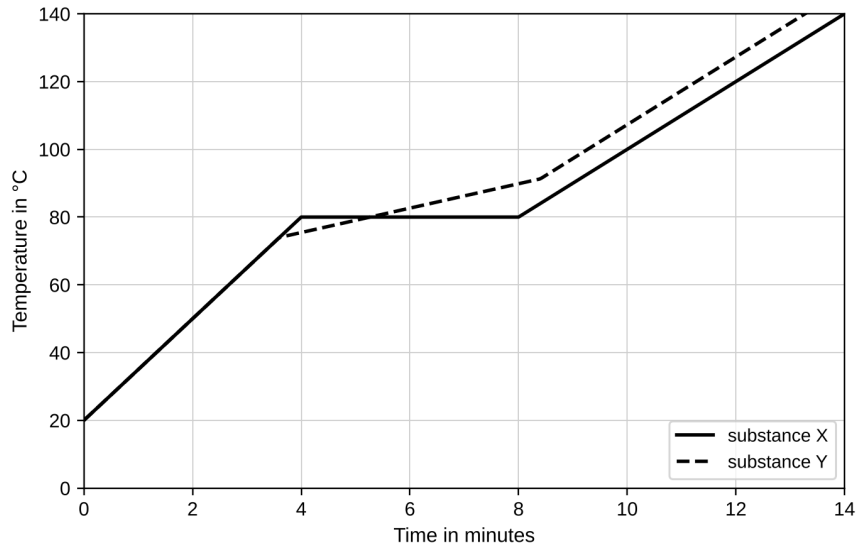
0 1 . 4 Give **two** examples of formulations. **[2 marks]**

0 1 . 5 Explain why the quantities of each component in a medicine must be carefully controlled. **[2 marks]**

0 1 . 6 A paint contains a pigment, a solvent, a binder and additives. Draw one line from each component to its job. **[3 marks]**

Component	**Job**
pigment	Holds the pigment on the surface when the paint dries
solvent	Gives the paint its colour
binder	Makes the paint runny so it can be spread

0 2 A student heats two solids, **X** and **Y**, and measures the temperature every minute. The graph shows the results.



0 2 . 1 Which solid is pure? Give **two** reasons for your answer. **[3 marks]**

0 2 . 2 What is the melting point of X? **[1 mark]**

0 2 . 3 For how many minutes is X melting? **[1 mark]**

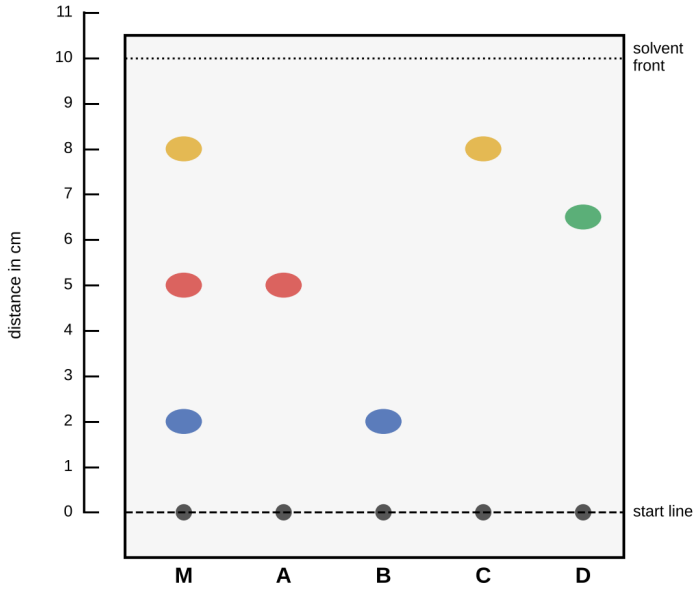
0 2 . 4 Describe the melting range of Y. **[1 mark]**

0 2 . 5 How can a scientist use the melting point to identify a pure substance? **[1 mark]**

0 2 . 6 Suggest what effect impurities have on the melting point of a pure solid. **[2 marks]**

0 2 . 7 Pure ethanol boils at 78 °C. A student's sample of ethanol begins boiling at 84 °C. What does this suggest? **[1 mark]**

0 3 A student uses paper chromatography to find out which dyes are in an ink, **M**. She compares it with four known dyes, **A**, **B**, **C** and **D**. The chromatogram is shown.



0 3 . 1 The start line is drawn in pencil. Explain why. **[1 mark]**

0 3 . 2 The start line must be above the level of the solvent in the beaker. Explain why. **[1 mark]**

0 3 . 3 Which dyes are in ink M? **[2 marks]**

0 3 . 4 Which dye is **not** in ink M? Explain how you know. **[2 marks]**

0 3 . 5 Is ink M a pure substance? Explain your answer. **[2 marks]**

0 3 . 6 Calculate the R_f value of dye A. **[2 marks]**

R_f = distance moved by substance ÷ distance moved by solvent

Answer = _____

0 3 . 7 Calculate the R_f value of dye D. Give your answer to 2 significant figures. **[2 marks]**

Answer = _____

0 3 . 8 Dye C has an R_f value of 0.80. Explain what this tells you about how strongly dye C is attracted to the paper compared with dye B (R_f = 0.20). **[3 marks]**

0 3 . 9 Explain why an R_f value can never be greater than 1. **[1 mark]**

0 4 This question is about tests for gases.

0 4 . 1 Complete the table to show the test and the positive result for each gas. **[4 marks]**

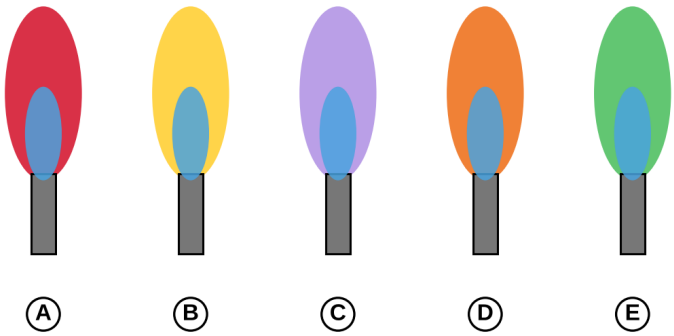
Gas	Test	Positive result
hydrogen		
oxygen		
carbon dioxide		
chlorine		

0 4 . 2 Write a balanced symbol equation for the reaction of carbon dioxide with limewater (calcium hydroxide solution). **[2 marks]**

0 4 . 3 A student bubbles carbon dioxide into limewater for several minutes. The cloudiness disappears. Suggest why. **[1 mark]**

0 4 . 4 A student collects a gas from a reaction. The gas turns limewater cloudy and extinguishes a lit splint. Name the gas. **[1 mark]**

0 5 A student carries out flame tests on five metal compounds. The diagram shows the flame colours, **A** to **E**.



0 5 . 1 The five metal ions are lithium, sodium, potassium, calcium and copper(II). Write the name of the ion in each flame. **[5 marks]**

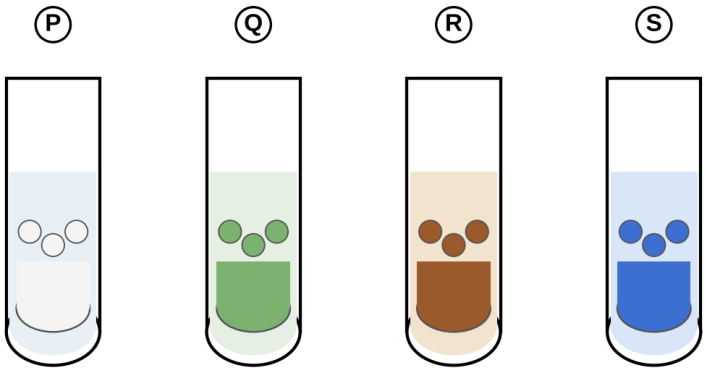
0 5 . 2 Describe how to carry out a flame test. **[3 marks]**

0 5 . 3 A student tests a mixture of sodium chloride and potassium chloride. The flame is bright yellow. Explain why the student cannot be sure that potassium is not present. **[2 marks]**

0 5 . 4 Give **one** reason why flame tests are better than chemical tests for identifying metal ions in a mixture, and **one** reason why they are not. **[2 marks]**

0 6 Sodium hydroxide solution is added to solutions containing different metal ions. The diagram shows the precipitates formed in four test tubes, **P**, **Q**, **R** and **S**.

P: white Q: green R: brown S: blue



0 6 . 1 Name the metal ion in each of test tubes **Q**, **R** and **S**. **[3 marks]**

0 6 . 2 Write an ionic equation, with state symbols, for the formation of the blue precipitate in S. **[2 marks]**

0 6 . 3 The ion in P could be aluminium, magnesium or calcium. A student adds excess sodium hydroxide solution to tube P and the precipitate does not dissolve. **[2 marks]**

Which ion is **not** in P? Explain your answer.

0 6 . 4 Describe how the student could use a flame test to show which of the two remaining ions is in P. **[2 marks]**

0 6 . 5 Iron(III) sulfate reacts with sodium hydroxide: $Fe_2(SO_4)_3$ + ___NaOH → ___$Fe(OH)_3$ + ___Na_2SO_4 **[2 marks]**
Balance the equation.

0 7 This question is about tests for negative ions.

0 7 . 1 Complete the table to show the test and the positive result for each ion. **[6 marks]**

Ion	Test	Positive result
carbonate, CO_3^{2-}		
sulfate, SO_4^{2-}		
chloride, Cl^-		

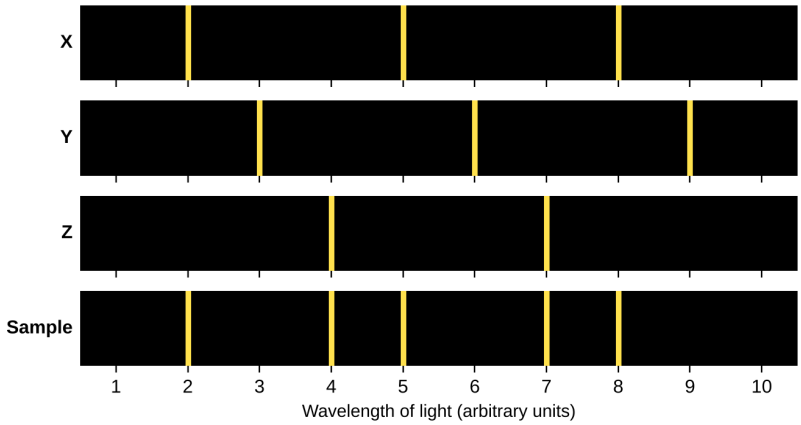
0 7 . 2 Name the precipitate formed when silver nitrate solution is added to a solution of sodium bromide, and give its colour. **[2 marks]**

0 7 . 3 In the test for halide ions the solution is first acidified with nitric acid. Suggest why. **[1 mark]**

0 7 . 4 Write an ionic equation for the reaction of barium ions with sulfate ions. **[2 marks]**

0 7 . 5 A solution gives a white precipitate with sodium hydroxide solution. The precipitate dissolves when more sodium hydroxide is added. When silver nitrate and nitric acid are added to the original solution a cream precipitate forms.
Name the compound in the solution. **[3 marks]**

0 8 A scientist analyses a sample of river water using flame emission spectroscopy. The diagram shows the line spectra of three metal ions, **X**, **Y** and **Z**, and of the sample.



0 8 . 1 Which metal ions are in the sample? Explain your answer. **[3 marks]**

0 8 . 2 Give **two** advantages of using instrumental methods rather than chemical tests. **[2 marks]**

0 8 . 3 Describe the main steps in flame emission spectroscopy. **[2 marks]**

0 8 . 4 A student says 'chemical tests are always better than instrumental methods because they cost less.' Evaluate this statement. **[2 marks]**

0 9 A student carries out tests on four unknown solutions, **W**, **X**, **Y** and **Z**. The results are shown in the table.

0 9 . 1 Identify the compound in each solution. **[8 marks]**

Solution	Test with sodium hydroxide	Test with acid and a reagent	Flame test
W	blue precipitate	dilute HCl + $BaCl_2$: white precipitate	—
X	green precipitate	dilute HNO_3 + $AgNO_3$: white precipitate	—
Y	brown precipitate	dilute HNO_3 + $AgNO_3$: cream precipitate	—
Z	no precipitate	dilute HCl: gas that turns limewater cloudy	yellow

1 0 Stretch questions (grade 7–9).

1 0 . 1 *A student has a colourless solution that contains either magnesium chloride or aluminium sulfate. **[6 marks]**

Describe the tests the student should carry out to find out which compound is in the solution. Include the results and the ionic equation for one of the precipitates.

1 0 . 2 *A scientist suspects a river is polluted with copper compounds from a mine. **[6 marks]**

Compare the use of chemical tests and instrumental methods to find out whether copper ions are present and how much there is.

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.8: Chemical analysis

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	a single element or compound / not mixed with any other substance	Do not accept 'nothing added' or 'natural'.	1	AO1 4.8.1.1
01.2	milk is a mixture ; of different substances (water, fats, proteins, sugars) ;		2	AO2 4.8.1.1
01.3	a mixture that has been designed as a useful product ; made by mixing components in carefully measured quantities ;		2	AO1 4.8.1.2
01.4	fuels / cleaning agents / paints / medicines / alloys / fertilisers / foods ; any second example ;	Any two. Do not accept 'plastic'.	2	AO1 4.8.1.2
01.5	so the medicine has the correct effect / works properly ; too much could be harmful / too little would not work ;		2	AO2 4.8.1.2
01.6	pigment — gives the colour ; solvent — makes the paint runny / dissolves the other components so it spreads ; binder — holds the pigment on the surface when the paint dries ;	Candidates may draw lines or write letters. One mark for each correct link.	3	AO1 4.8.1.2
02.1	X ; the temperature stays constant (at 80 °C) while it melts ; Y melts over a range of temperatures ;		3	AO3 4.8.1.1
02.2	80 °C		1	AO3 4.8.1.1
02.3	4 (minutes)	Allow 4 min from 4 to 8 minutes.	1	AO3 4.8.1.1
02.4	from about 74 °C to 91 °C (allow ± 3 °C)		1	AO3 4.8.1.1
02.5	compare with the (data book / known) melting point		1	AO2 4.8.1.1
02.6	lowers the melting point ; melts over a range of temperatures ;		2	AO2 4.8.1.1
02.7	the sample is impure / contains dissolved substances		1	AO3 4.8.1.1
03.1	pencil is insoluble / does not dissolve / will not run (ink would run with the solvent)		1	AO3 4.8.2.1
03.2	so the dyes do not dissolve into the solvent in the beaker		1	AO3 4.8.2.1
03.3	A, B and C ; (all three for 2 marks; two correct with none wrong for 1 mark) ;		2	AO3 4.8.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	D ; no spot in M at the same height as D ;		2	AO3 4.8.2.1
03.5	no ; it has separated into three spots / contains three dyes ;		2	AO3 4.8.2.1
03.6	5.0 ÷ 10.0 ; 0.50 / 0.5 ;	Allow ± 0.5 mm in the measurements. No unit: R_f is a ratio.	2	AO2 4.8.2.1
03.7	6.5 ÷ 10.0 ; 0.65 ;		2	AO2 4.8.2.1
03.8	dye C moved further up the paper ; dye C is more soluble in the solvent / spends more time in the mobile phase ; dye C is less strongly attracted to the paper (stationary phase) than dye B ;		3	AO3 4.8.2.1
03.9	a substance cannot travel further than the solvent front		1	AO3 4.8.2.1
04.1	hydrogen: lit splint ; squeaky pop ; oxygen: glowing splint ; relights ; carbon dioxide: limewater ; turns cloudy / milky ; chlorine: damp (blue) litmus paper ; bleached / turns white ;	One mark for each gas (test and result both correct). Allow 'turns red then white' for chlorine.	4	AO1 4.8.3.1
04.2	$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$; correct formulae and balanced ;	Allow $Ca(OH)_2(aq)$ as limewater. The white precipitate is calcium carbonate.	2	AO2 4.8.3.1
04.3	excess carbon dioxide forms soluble calcium hydrogencarbonate		1	AO3 4.8.3.1
04.4	carbon dioxide		1	AO2 4.8.3.1
05.1	A: lithium (crimson / red) ; B: sodium (yellow) ; C: potassium (lilac) ; D: calcium (orange-red / brick red) ; E: copper(II) (green / blue-green) ;	One mark for each correct ion.	5	AO1 4.8.3.2
05.2	clean a flame test loop / wire (dip in acid and heat in the flame) ; dip the loop into the sample (solid or solution) ; hold it in a (blue) Bunsen flame and observe the colour ;		3	AO1 4.8.3.2
05.3	sodium gives a strong yellow flame ; which masks the lilac of potassium ;		2	AO3 4.8.3.2
05.4	advantage: quick / simple / small sample ; disadvantage: colours can be masked in mixtures / difficult to distinguish similar colours ;	Any suitable pair.	2	AO3 4.8.3.2
06.1	Q: iron(II) / Fe^{2+} ; R: iron(III) / Fe^{3+} ; S: copper(II) / Cu^{2+} ;		3	AO1 4.8.3.3
06.2	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$; correct formulae / state symbols ;	One mark for the correct species, one mark for balance and state symbols.	2	AO2 4.8.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	aluminium ; aluminium hydroxide dissolves in excess sodium hydroxide (the others do not) ;		2	AO3 4.8.3.3
06.4	calcium gives an orange-red flame ; magnesium gives no flame colour ;		2	AO3 4.8.3.2
06.5	$Fe_2(SO_4)_3 + 6NaOH \rightarrow 2Fe(OH)_3 + 3Na_2SO_4$; all four numbers correct (1 mark for two correct) ;		2	AO2 4.8.3.3
07.1	carbonate: add dilute acid ; bubbles of gas that turns limewater cloudy ; sulfate: add dilute hydrochloric acid then barium chloride solution ; white precipitate ; chloride: add dilute nitric acid then silver nitrate solution ; white precipitate (bromide cream, iodide yellow) ;	One mark for each correct test and each correct result. Allow barium nitrate / dilute nitric acid for sulfate.	6	AO1 4.8.3.4
07.2	silver bromide ; cream ;		2	AO1 4.8.3.4
07.3	to remove carbonate / hydroxide ions that would also form a precipitate with silver nitrate		1	AO3 4.8.3.4
07.4	$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4$; state symbols (aq, aq, s) ;		2	AO2 4.8.3.4
07.5	aluminium ions ; bromide ions ; aluminium bromide ;		3	AO3 4.8.3.4
08.1	X and Z ; the lines in the sample match all the lines of X ; and all the lines of Z (none of Y's lines are present) ;		3	AO3 4.8.4
08.2	rapid ; accurate / sensitive (can detect very small amounts) ;	Allow 'only a small sample is needed' or 'can be automated'. Any two.	2	AO1 4.8.4
08.3	the sample is heated in a flame / ionised so it emits light ; the light is passed through a spectroscope and the line spectrum compared with known spectra ;		2	AO1 4.8.4
08.4	chemical tests are cheaper / need no special equipment (partly true) ; but are slower, less sensitive and less accurate, so instrumental methods are better for small amounts / mixtures ;		2	AO3 4.8.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.1	W: copper(II) sulfate ; (blue precipitate = Cu^{2+}; white precipitate with barium chloride = sulfate) ; X: iron(II) chloride ; (green precipitate = Fe^{2+}; white precipitate with silver nitrate = chloride) ; Y: iron(III) bromide ; (brown precipitate = Fe^{3+}; cream precipitate = bromide) ; Z: sodium carbonate ; (yellow flame = Na^{+}; gas turns limewater cloudy = carbonate) ;	2 marks for each solution: one mark for the positive ion and one mark for the negative ion.	8	AO3 4.8.3
10.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • add sodium hydroxide solution to a sample: a white precipitate forms with both • add excess sodium hydroxide: the precipitate dissolves for aluminium ions but stays for magnesium ions • ionic equation e.g. $Mg^{2+}(aq) + 2OH^{-}(aq) \rightarrow Mg(OH)_2(s)$ or $Al^{3+}(aq) + 3OH^{-}(aq) \rightarrow Al(OH)_3(s)$ • for the negative ion: add dilute hydrochloric acid then barium chloride: white precipitate means sulfate • or add dilute nitric acid then silver nitrate: white precipitate means chloride • conclusion links the results: dissolving white precipitate + sulfate test positive = aluminium sulfate		6	AO3 4.8.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • chemical test: add sodium hydroxide; a blue precipitate shows copper(II) ions • the test is quick, cheap, and needs simple equipment, so it can be done at the river • but it is not very sensitive: very small amounts may not give a visible precipitate • it only shows that copper ions are present; it does not give an accurate concentration • instrumental method (e.g. flame emission spectroscopy) is rapid, very sensitive and accurate; gives the amount of each ion • it can identify several ions at once but needs expensive equipment and trained staff; conclusion: use a chemical test first then an instrument for accurate data		6	AO3 4.8.4

Where marks are lost in Topic 4.8

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Pure vs mixture:** in chemistry 'pure' means a single substance only (one element or compound). Pure substances melt and boil at fixed temperatures; impure ones melt over a range, at a lower temperature.
- ☐ **Chromatography:** pencil start line, above the solvent. R_f = distance moved by substance ÷ distance moved by solvent front. Measure to the *centre* of each spot. R_f has no unit and is always less than 1.
- ☐ **Why dyes move different distances:** more soluble in the solvent and/or less attracted to the paper = moves further.
- ☐ **Gas tests:** H_2 squeaky pop; O_2 relights a glowing splint; CO_2 turns limewater milky; Cl_2 bleaches damp litmus paper. Say what the colour/result is, not just 'it changes'.
- ☐ **Flame tests:** Li crimson, Na yellow, K lilac, Ca orange-red, Cu green. Mixtures can mask colours.
- ☐ **Hydroxide precipitates:** Cu^{2+} blue, Fe^{2+} green, Fe^{3+} brown. Al^{3+}, Mg^{2+}, Ca^{2+} give white; only aluminium re-dissolves in excess NaOH.
- ☐ **Anion tests:** carbonate (acid, CO_2), sulfate (HCl + $BaCl_2$, white), halide (HNO_3 + $AgNO_3$: white Cl, cream Br, yellow I). The acid goes in *first* to remove carbonate.
- ☐ **Identifying unknowns:** name the positive ion *and* the negative ion, then the compound. Say which test gave each.
- ☐ **Instrumental methods:** faster, more accurate, more sensitive, small samples. Disadvantages: expensive, need training.